



Radiomics for clinical decision support in radiation oncology

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Abstract

Radiomics is a promising tool for the development of quantitative biomarkers to support clinical decision-making. It has been shown to improve the prediction of response to treatment and outcome in different settings, particularly in the field of radiation oncology by optimising the dose delivery solutions and reducing the rate of radiation-induced side effects, leading to a fully personalised approach.

Despite the promising results offered by radiomics at each of these stages, standardised methodologies, reproducibility and interpretability of results are still lacking, limiting the potential clinical impact of these tools.

In this review, we briefly describe the principles of radiomics and the most relevant applications of radiomics at each stage of cancer management in the framework of radiation oncology. Furthermore, the integration of radiomics into clinical decision support systems is analysed, defining the challenges and offering possible solutions for translating radiomics into a clinically applicable tool.

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Introduction

The advancement of radiological imaging in the last decade has introduced an innovative diagnostic approach, known as radiomics [1].

Imaging in oncology has been classically based on the evaluation of visual metrics (size or metabolic activity of the lesions) and is susceptible to subjective interpretation [2]. However, it has become gradually evident that radiological images contain quantitative data, that may not be detectable by visual assessment [2]. Essentially, radiomics is a process of extraction of these quantitative features from radiological images and conversion into large mineable data [3]. It is assumed that these data are indicative of the tumour's biological characteristics [4].

Radiotherapy (RT) is an important treatment modality in oncology and is received by approximately 50% of all cancer patients during their course of illness [5]. In recent years,

the efficacy of RT techniques has improved due to better treatment planning, imaging, novel irradiation and artificial intelligence (AI) technologies [6]. Precise tumour targeting and individualised treatment plans are essential for optimising outcomes while minimising side effects. In this context, radiomics provides a valuable tool to predict tumour behaviour, responses to treatment, and potential adverse events before they become clinically evident, allowing for treatment tailoring [7].

Clinical decision-making is a core component of the care of patients with cancer [8]. It is a very complex process that has to take simultaneously into account different knowledge domains, ranging from patients' factors to molecular characteristics of the disease. For example, tumour heterogeneity is subject to variations occurring at different levels (intra-tumour and inter-tumour), representing a substantial issue. Intra-tumour heterogeneity is the coexistence of cell populations with different phenotypes within a single lesion, while inter-tumour heterogeneity refers to different tumour subclones at different sites of disease [9,10]. Conventional imaging methods, based on gross tumour characteristics like size, morphology and glucose uptake, while

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useful, can barely depict the full extent of this heterogeneity, potentially leading to incomplete information and limited evidence to support the clinical decisions [9,10]. This limitation underscores the necessity for advanced approaches, such as radiomics, which can analyse the spatial and temporal complexity of tumour phenotypes [11].

Imaging is essential in supporting decision-making in clinical practice [12]. Tumours can be indeed examined in several ways, depending on the different imaging techniques. In general, computed tomography (CT) imaging is often used to acquire images with high spatial resolution and plays a vital role in all phases of cancer management, including staging, treatment planning, biopsy guidance for diagnosis, and treatment response evaluation [13,14]. The integration of positron emission tomography (PET) with CT (PET/CT) provides functional information, identifying metabolically active tumour regions [15], assuming a primary role in the assessment of lymph nodes and distant metastases. Magnetic resonance imaging (MRI) offers excellent soft-tissue contrast and functional information, thanks to dynamic contrast-enhanced (DCE) and diffusion-weighted imaging (DWI) [16]. DWI measures the restriction of the diffusion of hydrogen atoms in water molecules, reflecting high cellularity, and plays a fundamental role in oncological imaging for both staging and treatment response assessment. In addition, the apparent diffusion coefficient (ADC) value, a quantitative metric for diffusion restriction, can be calculated. DCE provides qualitative and quantitative information on tumour vascularisation and permeability. However, the inter- and intra-scanner reproducibility of ADC and DCE quantitative metrics are still insufficient for their clinical implementation [17].

This review aims to describe the applications of radiomics in RT, including its methodology and the potential predictive and prognostic value (Figure 1). Furthermore, we explore how radiomics can be integrated into clinical decision support systems, guiding the treatment decisions through data-driven approaches.

Principles of radiomics

Radiomics is based on the translation of medical images into multidimensional data that can be quantitatively analysed [18]. Preprocessing steps, such as image normalisation and resampling, are conducted prior to segmentation, which consists in delineating regions of interest (ROIs) either manually, semi-automatically or automatically. Manual segmentation is time-consuming and subject to inter- and intra-observer variability [19]. Semi-automatic and fully automatic segmentations — based on machine learning (ML) and deep learning (DL) algorithms — have advanced significantly, challenging the notion of limited generalisability [20–22]. Models such as nnU-net have demonstrated robustness across different imaging modalities, geometries, and dataset sizes without the need for dataset-specific manual adjustments [23,24]. These advancements have proven beneficial in radiation oncology

for delineating tumour volumes and assisting in adaptive radiation therapy [25,26].

The radiomics features extracted from the ROIs provide information about intra-tumour heterogeneity and complexity [27]. Shape-based features focus on the three-dimensional geometry of the tumour and provide information about its size, volume, and spatial relationship with surrounding structures. First-order statistics depict the distribution of voxel intensities within the delineated ROIs, interpreting metrics such as average intensity, variance, and skewness. Texture-based features describe the spatial patterns of voxel intensities. Higher-order features require to process imaging data into different mathematical domains to extract more information, such as the wavelet domain, which decomposes the image into components at different scales and orientations, enhancing the detection of textural features that may be imperceptible in the original imaging data [28]. Additionally, the Laplacian of Gaussian (LoG) filter serves as a critical component in this process, being an isotropic filter that is instrumental in edge detection across various image scales [29]. This multi-scale analysis can provide a detailed delineation of tumour edges and heterogeneity. Several software platforms and tools have been developed for feature extraction [11]. Pyradiomics, an open-source Python package, stands out for its comprehensive library of features and its commitment to becoming a reference standard for radiomic analysis [30]. It facilitates high-throughput extraction from medical imaging, promoting a consistent and reproducible approach to feature analysis. To address its initial configuration complexity, Pyradiomics has been now implemented as a plug-in in open-source software, providing an interactive graphical user interface (GUI) [11]. Another example is LifeX, which includes both segmentation and radiomics analysis tools and is recognised for its user-friendly interface and versatility in handling various image formats [31]. It is

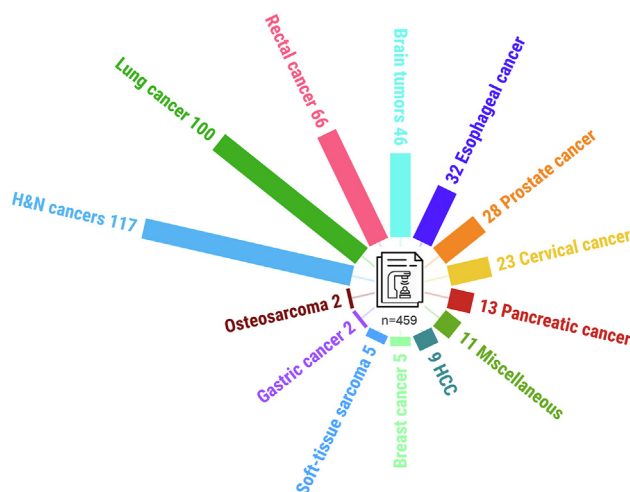


Fig 1. Radial bar chart showing the number of radiomics studies published in PubMed according to different organs treated with radiotherapy. At least 459 original articles were published until 22 August 2023.

H&N: head and neck; HCC: hepatocellular carcinoma.

particularly suitable for clinical researchers who may not be as experienced with coding. A significant effort within the radiomics community is geared towards standardising feature definitions and ensuring reproducibility across different platforms and datasets, such as the Image Biomarker Standardization Initiative (IBSI) [32].

After feature extraction, the next steps include feature selection and modelling, based on ML algorithms or statistical methods to identify the most predictive features. These selected features are then often integrated into predictive models that can provide prognostic or diagnostic information relevant to radiotherapy outcomes [27,33]. This workflow is achieved thanks to the concerted effort of a multidisciplinary team of professionals, consisting of radiologists, clinicians and computer/data scientists [34].

The evaluation of the reliability and quality of radiomics studies is essential for their interpretability and reproducibility. Indeed, several tools have been introduced for this purpose.

The radiomics quality score (RQS) takes into account different aspects of image acquisition, ROI delineation, feature extraction, and validation strategies [35]. A higher RQS value indicates a study that exhibits both methodological rigour and potential for clinical translation [36]. The Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) guidelines include a checklist to ensure the quality of predictive models [37]. Additionally, FAIR guiding principles ensure that the computational methods are sufficiently documented to allow for independent reproduction of the findings [38].

These tools establish a comprehensive evaluative framework that not only ensures the robustness of the study but also accelerates its clinical translation [7].

Applications of radiomics in RT

Predictive and prognostic implications

The most investigated applications of radiomics are its predictive and prognostic capabilities (Figures 2–3). The prediction of response to radiotherapy treatment and prognosis in terms of overall survival (OS) and disease-free (DFS) or progression-free survival (PFS), represent indeed the most studied outcomes (Table 1) [39].

The possibility to predict treatment response is of great interest as it could result in better-tailored treatment choices and the early identification of patients with poor response to treatment may give physicians time to modify the therapeutic approach [40].

Furthermore, this strategy may lead to toxicity reduction by de-escalating the dose in patients with favourable prognosis and avoiding overtreatments.

The analysis of the differences in radiomic features before, during or after treatment (the so-called “delta radiomics”) may be useful to explore innovative predictive solutions. Indeed, delta radiomics can potentially provide additional information on changes occurring in the early

phase of the treatment, resulting in better prediction of treatment response or prognostication. Interestingly, the association between delta radiomics and treatment response or prognosis has been studied in head and neck [41], lung [42], pancreatic [43], rectal cancers [44] and sarcoma [45].

The evidence on the use of delta radiomics using imaging acquired during RT treatment for image-guided radiotherapy (IGRT) purposes is still scarce despite its promising early results.

In a recent study, Chiloire *et al.* investigated the impact of dose escalation in the innovative framework of 0.35 T MRI-guided radiotherapy for rectal cancer patients predicted as poor responders thanks to the application of the “early-regression index (ERI)” during the treatment itself [46,47].

ERI compares gross tumour volume at the time of treatment simulation and throughout the whole neoadjuvant chemoradiotherapy treatment, representing a robust candidate biomarker for pathological complete response and DFS in locally advanced rectal cancer [46,48].

An ongoing clinical trial is applying this approach to support RT escalation [47]. The same group has also explored the application of such an approach in the setting of neoadjuvant chemoradiotherapy for cervical cancer [49].

Similarly, a radiomic model based on standard MRI combining gray-level co-occurrence-based features predicted two-year OS with AUC of 0.73 in the training set and 0.91 in the external validation set [50]. Gui *et al.* also observed that random forest models initialised with default parameters could predict pathological complete response after neoadjuvant chemoradiation therapy in locally advanced cervical cancer (AUC 0.80) [51].

A random survival forest model was found to be a useful prognostic predictor in patients undergoing induction chemotherapy plus concurrent chemoradiotherapy in a similar study on locally advanced nasopharyngeal carcinoma [52]. The incorporation of radiomic features in the model predicted a 5-year PFS with AUC of 0.89 in the training set and 0.84 in the internal validation set [52].

Furthermore, Gong *et al.* proposed a predictive model for local recurrence in oesophageal cancer combining CT-derived radiomics and clinical features producing a good performance in the internal (C-index 0.78) and external validation sets (C-index 0.76) [51].

Radiotherapy planning (segmentation and dose painting)

The contouring of the tumour and adjacent organs at risk (OARs) is a crucial step in RT treatment planning. This process involves the delineation of different tumour and target volumes: the gross tumour volume (GTV), clinical target volume (CTV) and the planning target volume (PTV). Radiomics has the potential to guide the automatic segmentation of the tumour [53–55].

Segmentation is a burdensome procedure for the radiation oncologist and its automatisisation may improve the inter- and intra-operator reproducibility and remarkably reduce the time [56].

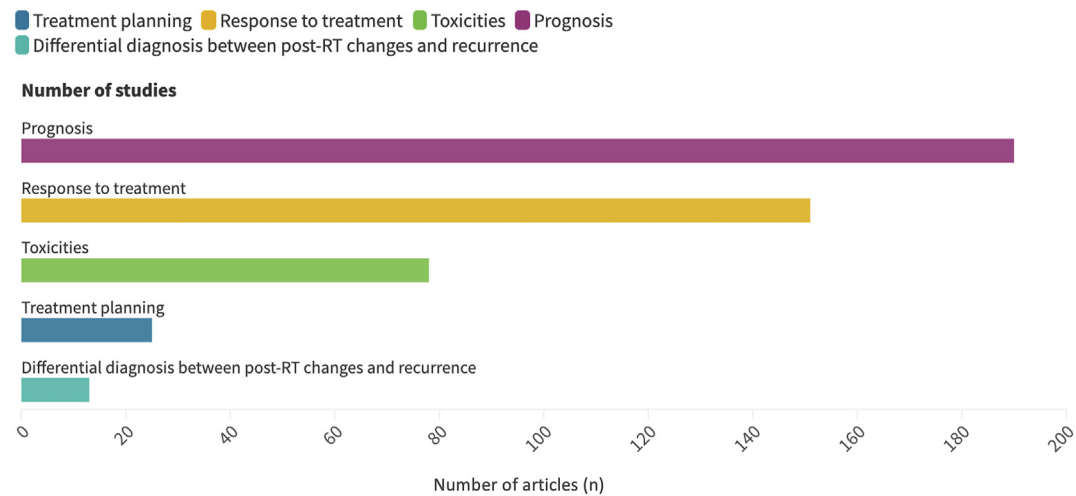


Fig 2. Bar chart illustrating the number of radiomics studies according to the field of application in radiation oncology (*until 22 August 2023). Most research focused on the prediction of prognosis (190 studies, 41.4%) and response after radiotherapy treatment (151 studies, 32.9%). The prediction of radiotherapy-induced toxicity was investigated in 71 studies (15.5%).

Besides standard diagnostic imaging, radiomics may have a role in guiding the identification of therapy volumes, opening a new perspective for the segmentation and prescription process.

In a study carried out by Shiradkar *et al.*, a radiomic classifier was trained for the delineation of prostate cancer and was externally validated. Radiomics-detected volumes were compared with those manually contoured by experienced radiologists. In particular, treatment plans generated from radiomics-detected volumes demonstrated significant

dose reductions in OARs and the chance to escalate the dose to the dominant intraprostatic lesions [57].

Radiomics allows for a detailed, non-invasive, and three-dimensional characterisation of the tumour phenotype, which contributes significantly to the tailoring of radiation doses [58]. Multiparametric functional imaging, including DCE- and DWI-MRI, and FDG PET/CT, has been employed to explore changes in tumour heterogeneity patterns and their correlation with radiation therapy response in cervical cancer [59]. In the clinical routine, while target volumes

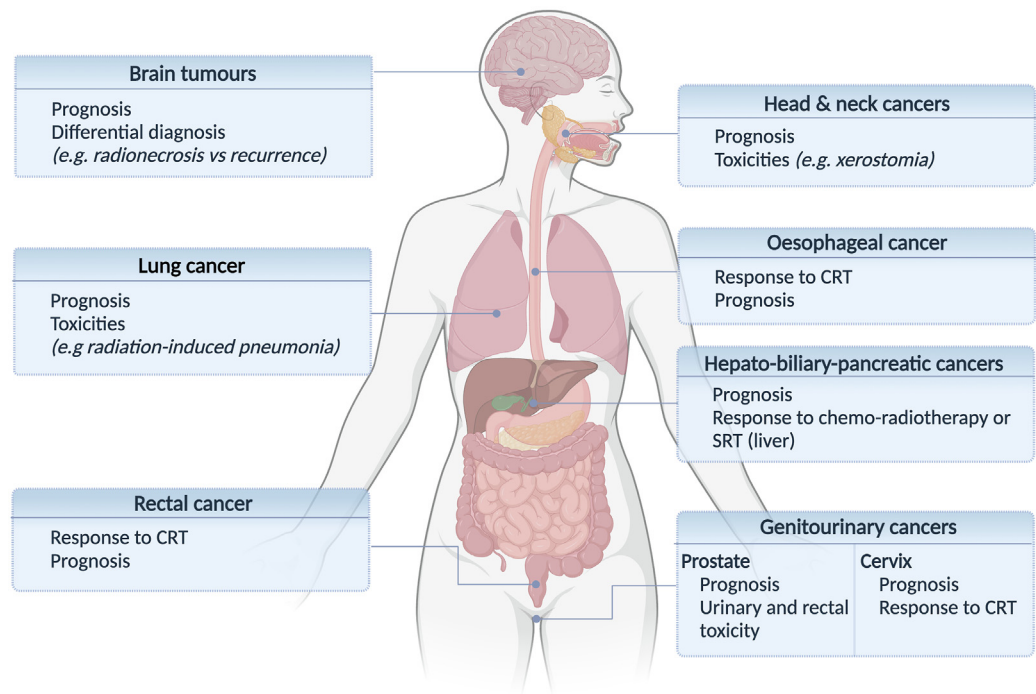


Fig 3. Illustration of the most frequently studied outcomes in radiomics research applied to radiotherapy. The outcomes are categorized by anatomical region (created with [Biorender.com](#)). CRT: chemoradiotherapy; SRT: stereotactic radiation therapy.

Table 1

Summary of radiomics applications in radiation oncology across different cancer types, highlighting key topics and outcomes from selected studies

Organ/system	Applications of radiomics	Outcome	Reference no.
Brain tumours	Differential diagnosis	Differentiation between recurrence and radiation-induced changes using radiomic signatures.	[78,79,81,83]
Head and Neck cancers	Prognosis	Prediction of RFS, PFS and OS using clinical and radiomics features	[7,52,91]
	Prediction of toxicities	Prediction of xerostomia after RT	[66–68]
	Response to treatment	MRI radiomics model and delta radiomics model to predict response to CRT	[41]
	Treatment planning	PET radiomics-based segmentation compared with manual segmentation	[53]
Lung cancer	Toxicities	Prediction of RP using treatment-related images, studies on dose reduction and complication prediction.	[69,70,72–75]
	Prognosis	Radiomics and delta radiomics for predicting OS and local recurrence	[42,82,89]
	Differential diagnosis	Differentiation between treatment response and RP	[71] [80],
	Treatment planning	Studies on radiomics for tailoring RT dose delivery and planning	[54,63]
Oesophageal cancer	Response to treatment	CT- and PET/CT-based radiomics for predicting response to CRT	[93,94]
	Prediction of prognosis	Predictive model for local recurrence-free survival using CT-based radiomics	[88]
Hepato-biliary-pancreatic cancers	Prediction of prognosis	Delta radiomics for prediction of local control in pancreatic cancer	[43]
	Treatment planning	PET radiomics-based segmentation of pancreatic cancer compared with manual segmentation	[53]
Rectal cancer	Response to treatment	Radiomics and delta radiomics models based on MRI or CT for predicting response to CRT	[44,87,92,96]
Cervical cancer	Response to treatment	Predictive MRI- and PET/CT-radiomics models for response to CRT	[51,59]
Prostate cancer	Prognosis	MRI-based prediction of DFS, OS or OFR	[50,90]
	Treatment planning	Radiomics-based segmentation and prediction models for RT planning.	[55,57,64]
	Prediction of toxicities	Prediction of GI and urinary toxicities	[76]
	Prognosis	Prediction of biochemical recurrence using MRI-based-radiomics	[95]

RFS: recurrence-free survival; PFS: progression-free survival; OS: overall survival; RT: radiation therapy; CRT: chemo-radiotherapy; RP: radiation-induced pneumonia; DFS: disease-free survival; OFR: out-of-field recurrence; GI: gastro-intestinal.

typically receive uniform doses of radiation, the possibility for a radiomics model to describe tumour heterogeneity paves the way towards biologically based dose painting [60,61]. This is achieved by providing a quantitative assessment of tumour heterogeneity, which can guide the differentiation of radiation doses across sub-volumes with different biological behaviours (i.e. sub-volumes with high risk of recurrence) in head and neck [62], lung [63], prostate [64] and cervical cancers [39,65].

Prediction of radiation-induced toxicity

Another significant field of investigation is the prediction of radiation-induced toxicity, using treatment-related images.

One of the most studied applications is the prediction of xerostomia in head and neck cancers undergoing RT

[66–68]. Another common radiation-induced toxicity is radiation pneumonitis (RP), which can be observed in patients treated with RT in the thoracic site [69–72]. Despite the large amount of data previously considered for the prediction of RP, findings are still inconclusive due to different factors, such as small sample size, inappropriate statistical evaluation or significant heterogeneity of the images [73,74]. Krafft *et al.* demonstrated that the addition of radiomic features improved the performance of a predictive model based on clinical and dosimetric data (AUC 0.68 vs 0.51) [74]. Similarly, Kawahara *et al.* reported the ability of a developed multi-region radiomics analysis to predict grade ≥ 2 RP and indicated variation and local intensity roughness, as principal features for this outcome [75]. However, current RP models are not able to adequately evaluate baseline differences in radiosensitivity and lung function [74].

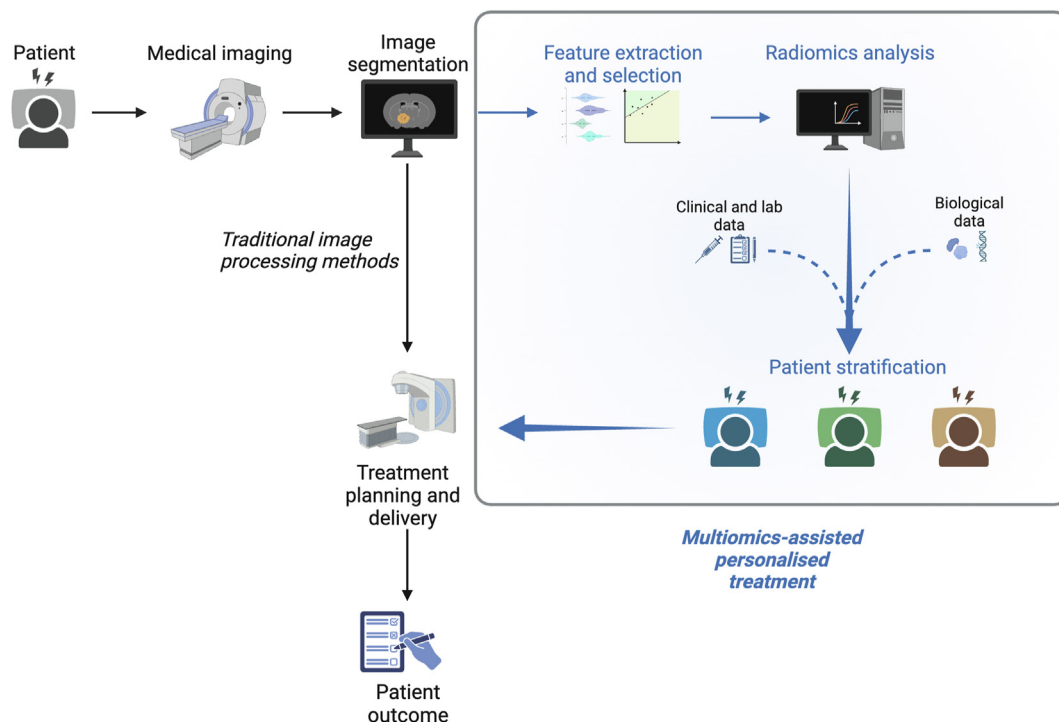


Fig 4. The workflow diagram depicts the journey from patient imaging to multiomics-assisted personalized treatment in radiation therapy, showing the integration of radiomics analysis and patients' data for optimized patients' stratification and treatment personalisation (created with [Biorender.com](https://biorender.com)).

In a recent study, a CT-based delta radiomics model accurately predicted severe acute RP in patients with oesophageal cancer [71].

Other common toxicities have been explored, such as in a model integrating both radiomic and clinical/dosimetric features predicting urinary (AUC 0.77) and gastrointestinal complications (AUC 0.71) [76].

In a similar study, the addition of 3D dosimetric texture analysis features (i.e. bladder and rectal texture analysis) to the predictive modelling of gastrointestinal and genitourinary toxicities in prostate cancer patients, was shown to improve predictive performance, especially for gastrointestinal toxicity which was statistically significant [77].

Differential diagnosis between tumour recurrence and radiation-induced changes

Differentiating therapy-induced modifications from recurrence/persistence of disease is a major need after radiation treatment, especially for lung cancer and brain cancer patients [36,78–81].

A radiomic signature demonstrated good performance (AUC 0.85) in differentiating between tumour recurrence and post-RT changes in patients irradiated for lung cancer [82], while a predictive model based on MR-derived features showed good performance with AUC 0.81 in differentiating radionecrosis from progressive disease after stereotactic radiotherapy for brain lesions [83].

Integration of radiomics into clinical decision support systems

The integration of radiomics into clinical decision support systems can provide the clinicians with data-driven insights to improve decision-making and tailor treatment, allowing for the precise delineation of tumour margins and identification of intra-tumour heterogeneity, which is critical for targeted therapy and dose painting in radiation therapy (Figure 4) [84]. Studies have demonstrated that radiomic features integrated into clinical models can improve the accuracy of treatment planning, offering superior delineation for radiation targeting compared to conventional imaging alone [85,86]. Furthermore, radiomics demonstrated significant predictive ability of prognosis [87–91] and response to treatment [92–94]. This is particularly valuable in radiation oncology. For example, radiomic signatures and nomograms derived from pre-treatment imaging can predict pathological response to neoadjuvant treatments and to stratify patients according to their risk or recurrence, allowing oncologists to customise treatment protocols – such as deciding between aggressive chemo-radiotherapy for high-risk patients or choosing less intensive treatments for low-risk groups [89,90,95]. In a retrospective study performed by Guo *et al.*, a 27-gene pair signature was identified with the ability to robustly predict pathological response to neoadjuvant chemoradiation in locally advanced rectal cancer patients. This 27-gene pair signature was developed through the

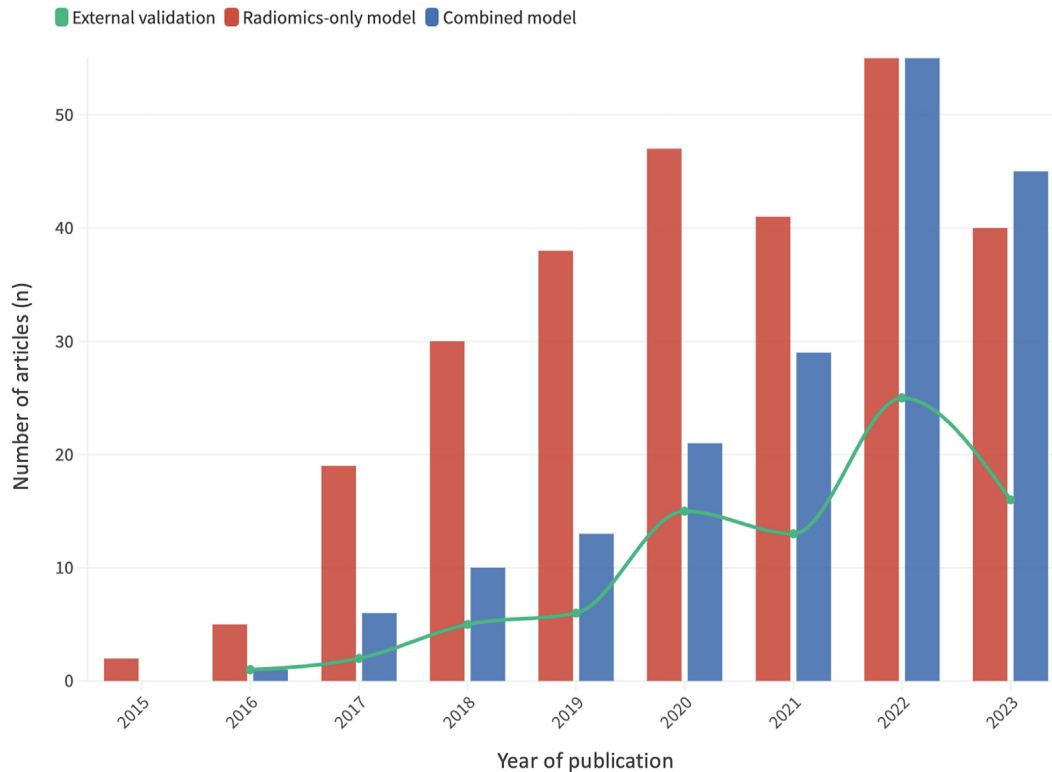


Fig 5. Column chart displaying the number of radiomics studies applied to radiotherapy up to 22 August 2023. The chart reveals a consistent increase in the number of models combining radiomic data with others (in blue) as compared to radiomic-only models (in red). In 2022, radiomics-only models were equal to those combined ($n = 55$). So far in 2023, the number of combined models ($n = 45$) exceeds that of radiomics-only models ($n = 40$). The green line indicates the number of studies with external validation.

screening of pre-treatment biopsies for gene pairs, whose relative expression levels correlated with pathological response, using gene expression analysis. This signature showed good predictive performance with an accuracy of 92.86% and an AUC of 0.95 in the training cohort, and accuracies of 89.13% and 90.91% with AUCs of 0.95 and 0.91 in the two independent validation cohorts [96]. In a more recent study, a model based on 142 collagen features extracted from the multiphoton image of pre-treatment biopsies, was developed and validated to investigate the association between collagen features and pathological complete response in locally advanced cancer patients. The performance of the nomogram was measured with respect to its discrimination, calibration, and clinical utility. This nomogram also demonstrated good discrimination in both primary and validation cohorts with AUCs of 0.842 and 0.836 respectively. In addition, decision curve analysis (DCA) and clinical impact curves (CICs) further demonstrated the predictive ability and clinical usefulness of the nomogram in identifying patients with potential pathological complete response. These results suggest the significant association between the collagen signature and pathological complete response and its usefulness in predicting pathologic complete response in locally advanced rectal cancer patients before neoadjuvant chemo-radiotherapy [97].

Radiomics can enhance scheduling optimisation, potentially spreading out treatment sessions for slow-growing tumours to minimize patient burden or intensifying the schedule for aggressive malignancies [98]. The possible prediction of radiation-induced toxicities, such as xerostomia or pneumonitis, permits the use of preemptive strategies, such as the use of protective agents, altered fractionation schedules, or proactive symptom management plans, to mitigate these effects, thereby improving the quality of life of the patients [70–72,99] (Figure 5).

The incorporation of radiomics into radiation oncology is driven by the complexity and multidimensionality of the radiomic data, including hundreds to thousands of features reflecting tumour heterogeneity and habitats; hence, it is crucial to prevent the detection of random significant results [100]. The use of AI algorithms can automatise the analysis process, overcoming the limitations of manual selection and interpretation, which is often time-consuming and subject to error [101]. The real-time nature of the decision support systems can have significant impact on treatment strategies, permitting to promptly change dose distribution or treatment regimens as appropriate, facilitating adaptive radiation therapy that responds to changes in tumour size, shape, and biology [102,103].

Challenges and limitations

The incorporation of radiomics into clinical decision support systems is not without its challenges [104]. A major issue arises from the lack of standardisation across different imaging modalities and processing techniques. This variability raises issues in radiation oncology studies, where precision in treatment planning is predominant, and yet, manual contouring remains the standard, introducing variability and potential bias in feature stability. This makes it challenging to compare and generalise radiomic features, potentially resulting in non-reproducible outcomes [32]. On the other hand, the selection of features that are most predictive in a specific clinical scenario can lead to “over-fitting” of the model. To address this, multicentric validation is a necessary requirement to guarantee the reproducibility of the models [105,106].

Another challenge faced by these integrated models is ensuring data privacy and security. Accessing large datasets that can include sensitive information of the patients makes it critical to preserve confidentiality while still allowing effective utilisation of these data [107,108].

Finally, the interpretability of ML models is an additional issue [109]. Especially the “black box” nature of DL-based algorithms can obstacle the comprehending of the rationale behind a specific decision, slowing the clinical translation in radiation oncology [110].

Future directions and perspectives

Despite the promising evidence in the current literature, the future development of radiomics should be critically analysed. The continuous advancement of AI algorithms and computational platforms holds the potential to enhance the accuracy as well as the robustness of integrated decision support systems, leading to the development of more personalized and adaptive treatment plans [111]. AI applications in radiation oncology could improve the precision and quality of radiation therapy by optimising the planning and delivery processes based on complex radiomic data [112].

Another field of interest is the potential real-time adaptation of radiotherapy regimens [113]. Prospective advancements could offer instantaneous data analytics, permitting the dynamic adjustment of treatment plan, while monitoring radiomic changes happening in the tumour environment and surrounding tissues.

The integration across multi-modal imaging – including morphologic (CT and MRI) and functional (PET and MRI) data – into a unified decision support system allows for a more holistic assessment. As radiomics continues to develop, there is a growing interest towards the “multi-omics” approach, integrating radiomic data principally with genomic information [114]. This combination can potentially provide a more comprehensive understanding of tumour biology by merging different sets of data.

Conclusions

Radiomics combines medical imaging and data science, promising significant advancement for transformative changes in treatment planning and prognosis of cancer patients [35]. This review explored radiomics applications, highlighting its potential utility in managing cancer patients treated with radiation therapy.

Multi-omics data incorporation into clinical decision support systems can represent a significant milestone. The number of predictive models integrating radiomics with clinical data has progressively increased. To improve the clinical relevance of these models and ensure their complete application in prospective clinical trials, rigorous methodological integrity is required. It is essential to conduct research to identify radiomic signatures that preserve their predictive ability regardless of the variables in the acquisition protocols. For this purpose, validation on external heterogeneous datasets is crucial. Other concerns regard the privacy and security of the clinical data. Indeed, the external validation of integrated predictive models necessitates implementing preventive measures for the safety and privacy of this data.

Henceforth, with continuous progress in AI and computational technologies, it is likely that decision support systems will become more precise and robust. Given the prospective precision of these tools, a real-time adaption of treatment plans through the integration of multi-omics data could represent a ground-breaking improvement in cancer patient care.

Statement of search strategies used and sources of information

This overview was conducted by performing a search of the literature, using the terms: “radiomics” or “radiomic” AND “radiation oncology” or “radiotherapy”, “computed tomography” or “CT”, “magnetic resonance imaging” or “MRI”, “ultrasound” or “US”, “PET/CT”. Relevant studies were reviewed and used as appropriate.

Author Contributions

- 1 Guarantor of integrity of the entire study: Evis Sala
- 2 Study concepts and design: Evis Sala, Luca Boldrini, Luca Russo
- 3 Literature research: Silvia Bottazzi, Dieprie Charles-Davies
- 4 Clinical studies: N/A
- 5 Experimental studies/data analysis: N/A
- 6 Statistical analysis: N/A
- 7 Manuscript preparation: Luca Russo
- 8 Manuscript editing: Luca Boldrini, Dieprie Charles-Davies

Conflict of Interest

The authors declare no conflict of interest.

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